

Applying abstraction

Revise 360

Worksheet for the 360° experience: Model design lab

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Model design lab. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Apply abstraction to a new problem.
- Identify inputs, processing and outputs.
- Refine a model when requirements change.

Starter: before you put the headset on

A program compares two places north or south. Does it need latitude, shop names or building colours?

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Key terminology

Input

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Output

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Define the question

Turn right

Key facts in your own words:

Challenge: Why define the output before selecting data?

Which input supports a north/south comparison?

The required output helps determine...

2 Signed coordinates

Turn right

Key facts in your own words:

Challenge: Which is further north: -8 or -21? Explain.

Which latitude is further north?

Which is further north: -15 or -4?

3 Equality matters

Turn around

Key facts in your own words:

Challenge: Why is an equality case needed?

A = 12 and B = 12. What should be output?

Which comparison checks equal values?

4 Inputs and processing

Turn around

Key facts in your own words:

Challenge: Classify multiplying distance by 2.

Classify each part of the fare model.

Categories: Input / Process / Output

Item	Category
Distance entered	
Multiply by 2	
Fare displayed	

A displayed fare is...

5 Change the requirement

Turn left

Key facts in your own words:

Challenge: What extra coordinate supports east/west comparison?

A fare now depends on passenger age. What changes?

Which input supports an age discount?

6 Test the simplified model

Turn left

Key facts in your own words:

Challenge: Why test equal and negative latitudes?

Order the model-development steps.

Put these steps in order: Check known examples / Write the processing rules / Choose relevant inputs / State the required answer

Which case checks equality?

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

A fare is distance * 2. Distance is 7. What is output?

Match each improvement to its purpose.

Match each item to a description from the list below.

Apply an age discount / Reduce irrelevant detail / Handle equal values

Item	Matching description
Add an equality branch	
Remove wall colours	
Add age	

Your score

Copy it from the score screen as soon as it appears.

_____ / 21

Key question

Can a simplified model give a correct answer?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Identify input, process and output for a fare calculated as distance multiplied by 2. [3]

b) Explain why comparing only positive latitudes is insufficient testing. [2]

c) A model compares only north/south locations. Explain a refinement needed for equal latitudes. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Apply abstraction to a new problem.			
Identify inputs, processing and outputs.			
Refine a model when requirements change.			